

METIMMOX-1 Cost-Effectiveness Analysis Report

Biomarker-Guided Strategies for Selecting Immunotherapy Candidates in Metastatic MSS/pMMR Colorectal Cancer

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1 Overview

This cost-effectiveness analysis evaluates three biomarker-guided strategies (CRP, TLR, and TMB/BRAF) for selecting metastatic microsatellite stable (MSS)/proficient mismatch repair (pMMR) colorectal cancer patients for immunotherapy versus the standard of care strategy of chemotherapy alone.

This report contains:

- **Strategy Descriptions:** Overview of the four treatment strategies evaluated
- **Base Case Analysis:** Cost-effectiveness results with incremental cost-effectiveness ratios (ICERs) and cost-effectiveness plane
- **Deterministic Sensitivity Analysis:** Tornado diagrams showing parameter impact on each strategy
- **Probabilistic Sensitivity Analysis:** Monte Carlo simulation results including scatter-plot and cost-effectiveness acceptability curves

2 Strategies Evaluated

Table 1: Treatment Strategies Evaluated in the Cost-Effectiveness Analysis

Strategy	Description	Biomarker Preva- lence	Patients (n)
Standard of Care	Standard of care - All patients receive only FLOX chemotherapy	100.0%	36
CRP-guided Strategy	C-reactive protein with cut-off of <5 for biomarker-positive status. If CRP-positive: alternating two cycles each of FLOX (chemotherapy) and nivolumab (anti-PD1 immunotherapy); if CRP-negative: chemotherapy only	33.8%	45
TLR-guided Strategy	Tumor Lesion Reduction with cut-off of $\geq 10\%$ for biomarker-positive status. If TLR-positive: alternating two cycles each of FLOX (chemotherapy) and nivolumab (anti-PD1 immunotherapy); if TLR-negative: chemotherapy only	64.7%	26
TMB/BRAF- guided Strategy	Combined biomarker: either Tumor Mutation Burden ≥ 9 or BRAF V600 mutation positive (both from next-generation sequencing). If TMB/BRAF-positive: alternating two cycles each of FLOX (chemotherapy) and nivolumab (anti-PD1 immunotherapy); if TMB/BRAF-negative: chemotherapy only	44.9%	34

3 Base Case Analysis

Table 2: Base Case Cost-Effectiveness Results with ICERs

Strategy	Cost	Effect	Inc_Cost	Inc_Effect	ICER	Status
control	\$21,827	1.892	NA	NA	NA	ND
crp	\$56,299	2.014	34471.96	0.121884	\$282,826	ND
tmb_braf	\$68,664	1.944	NA	NA	NA	D
tlr	\$77,574	1.359	NA	NA	NA	D

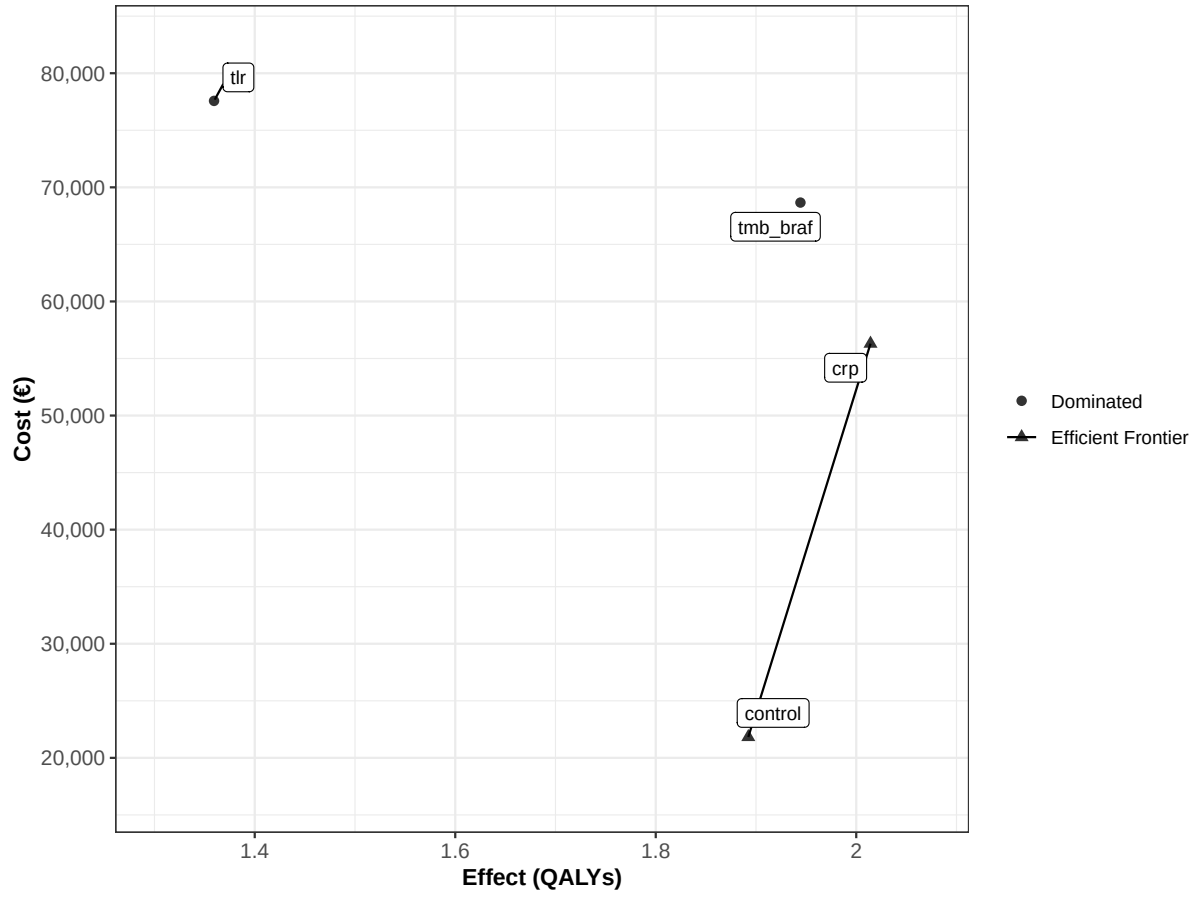


Figure 1: Cost-Effectiveness Plane with Efficiency Frontier

4 Deterministic Sensitivity Analysis

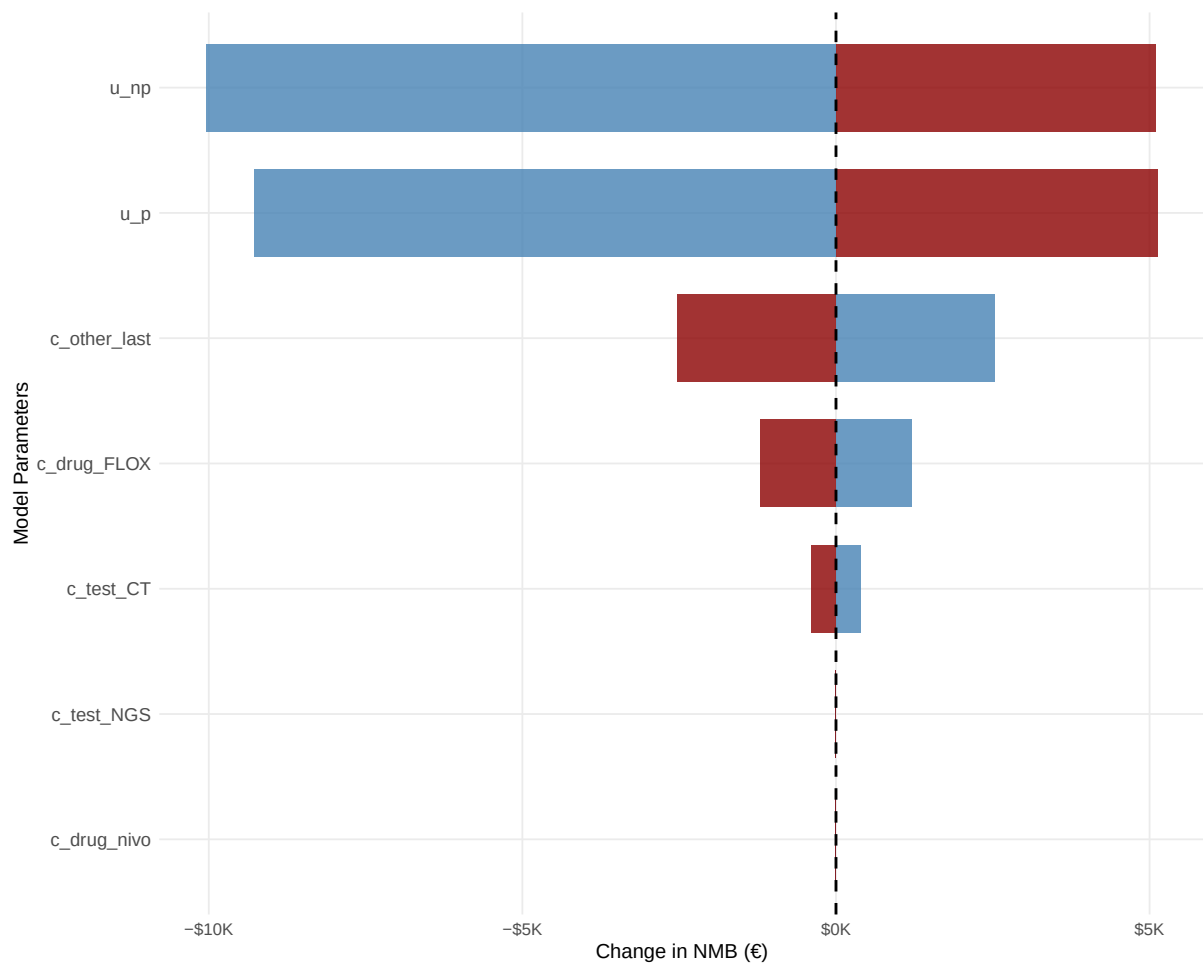


Figure 2: Tornado Diagram - Standard of Care One-Way Sensitivity Analysis

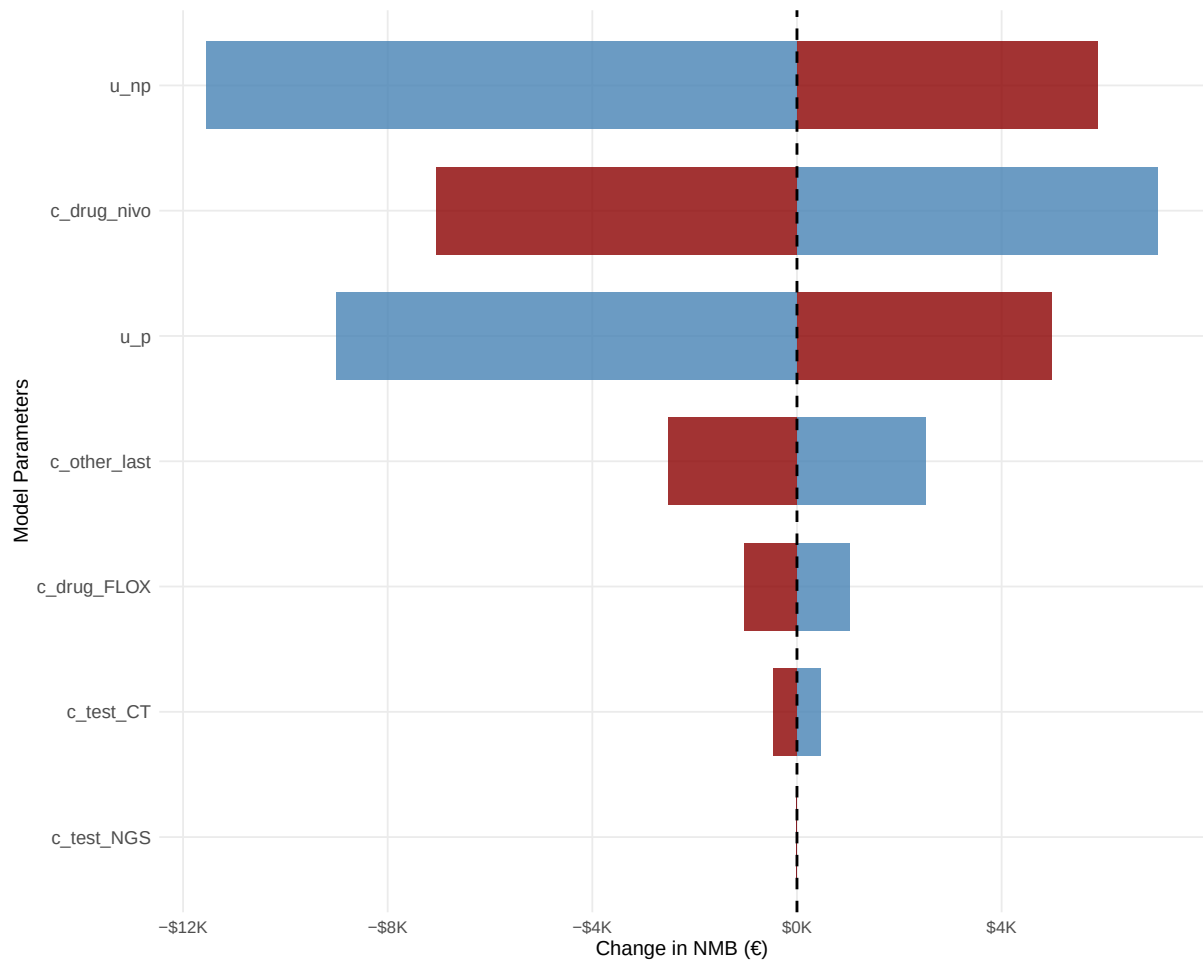


Figure 3: Tornado Diagram - CRP-guided Strategy One-Way Sensitivity Analysis

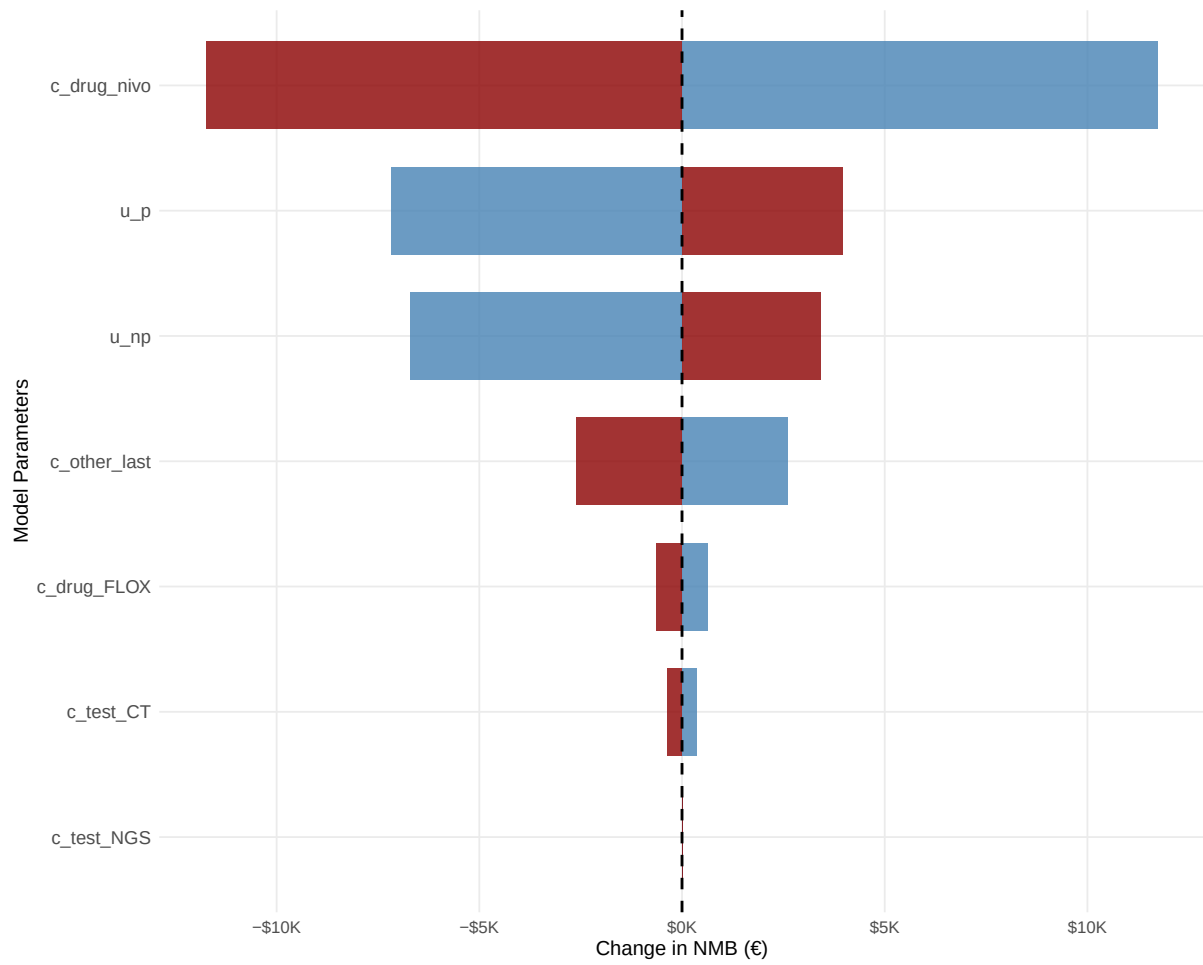


Figure 4: Tornado Diagram - TLR-guided Strategy One-Way Sensitivity Analysis

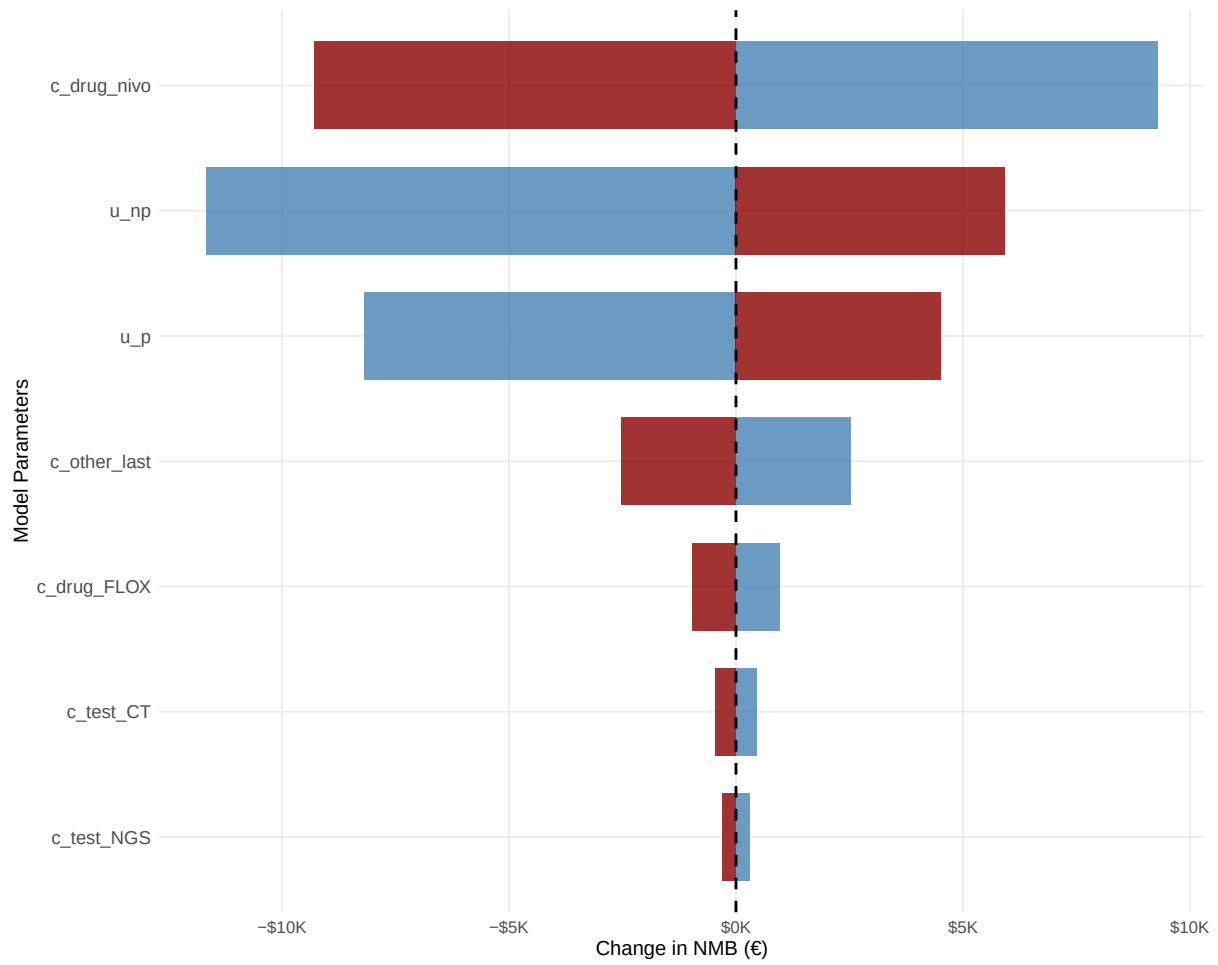


Figure 5: Tornado Diagram - TMB/BRAF-guided Strategy One-Way Sensitivity Analysis

Table 3: Parameter Impact Summary - Most Influential Parameters by Strategy

Rank	Parameter	Impact on NMB (€)
Standard Care		
1	Utility (No Progression)	\$15,131
2	Utility (Progression)	\$14,392
3	End-of-Life Care Cost	\$5,068
CRP-guided		
1	Utility (No Progression)	\$17,401
2	Nivolumab Cost	\$14,084
3	Utility (Progression)	\$13,987
TLR-guided		
1	Nivolumab Cost	\$23,444
2	Utility (Progression)	\$11,127
3	Utility (No Progression)	\$10,105
TMB/BRAF-guided		
1	Nivolumab Cost	\$18,559
2	Utility (No Progression)	\$17,584
3	Utility (Progression)	\$12,692

5 Probabilistic Sensitivity Analysis

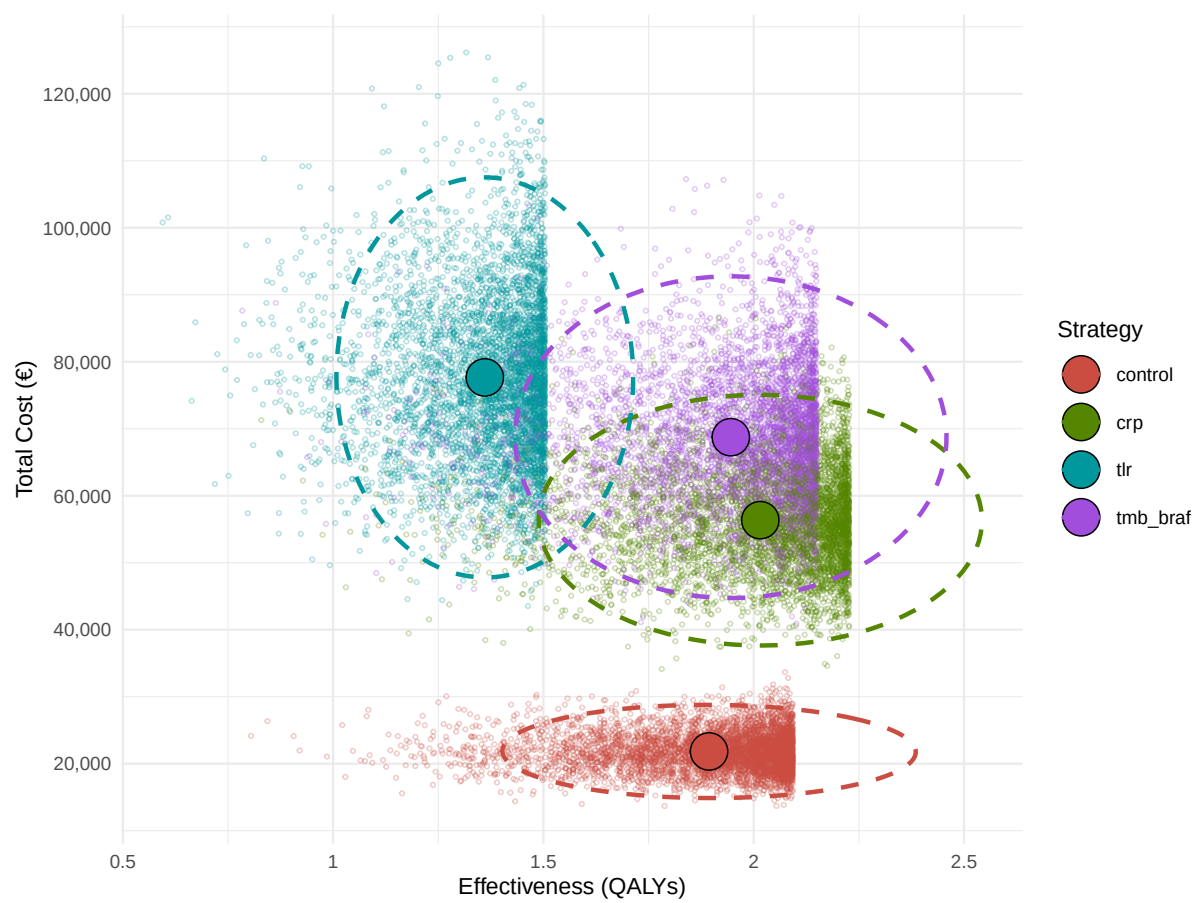


Figure 6: Cost-effectiveness Scatterplot

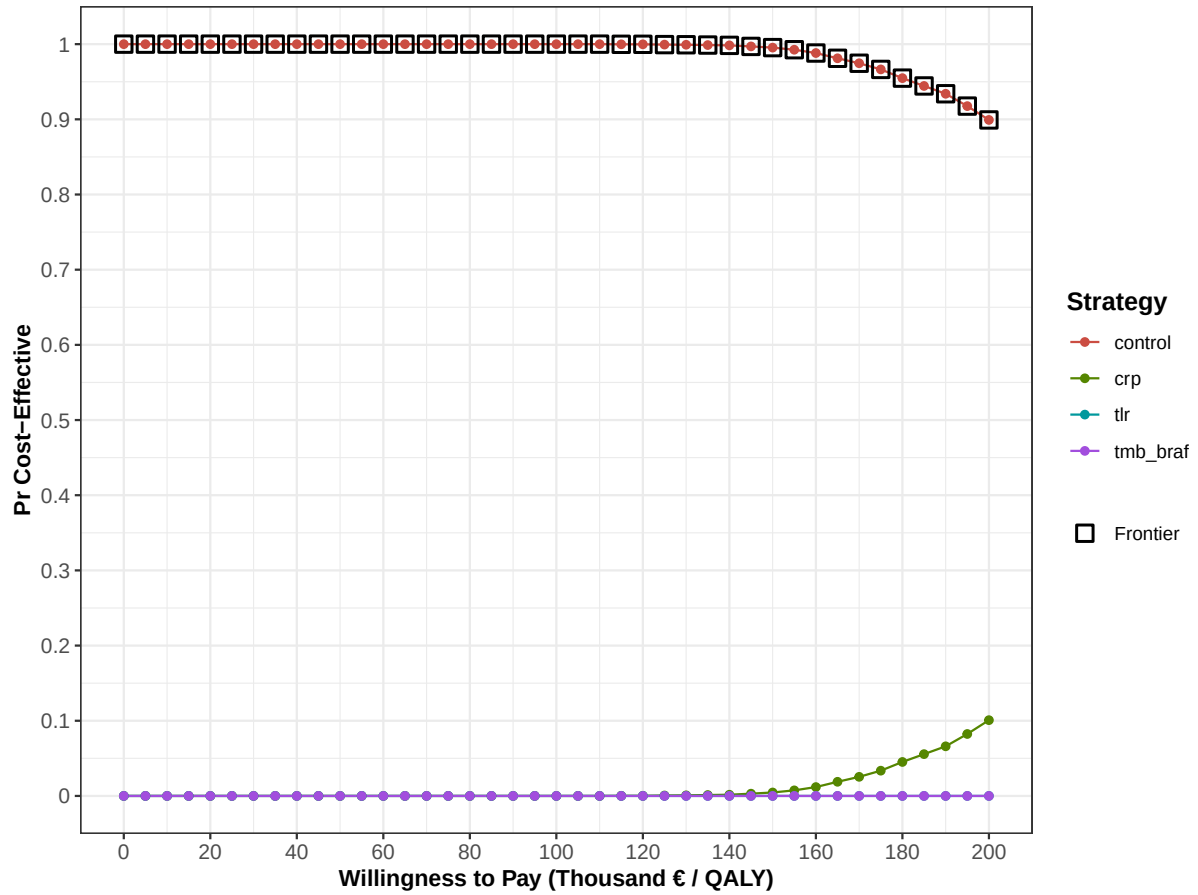


Figure 7: Cost-Effectiveness Acceptability Curve

6 Other Reports

I am working on the following other reports:

- **Input Parameters Report:** Comprehensive documentation of all model input parameters including clinical effectiveness data, cost parameters, and utility values with their respective sources and uncertainty distributions.
- **Parametric Survival Analysis Report:** Detailed documentation of the parametric survival modeling process, including goodness-of-fit statistics for all tested distributions, model selection criteria, and validation of the chosen survival models for both overall survival and progression-free survival endpoints.
- **One-Way Sensitivity Analysis (OWSA) Report:** Extended sensitivity analysis providing detailed examination of each individual parameter's impact on model outcomes, including threshold analyses and parameter sensitivity rankings across all strategies.

Report completed on: 2025-10-07

Repository: ben-geisler/METIMMOX-1

Report version: 1.0